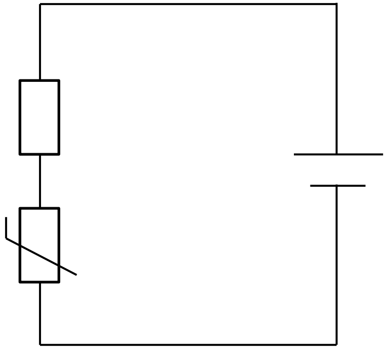
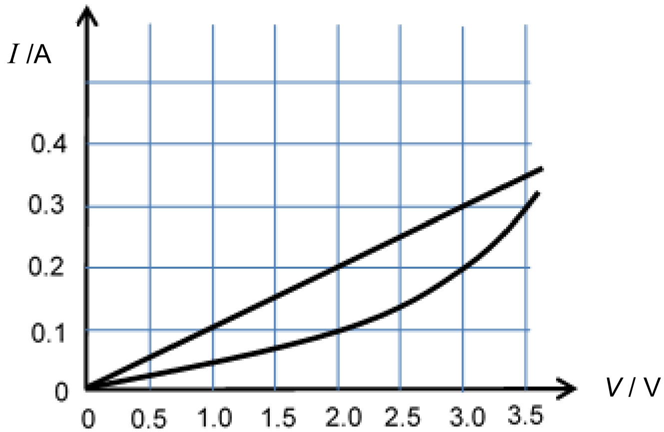
21 A 10 Ω resistor and a thermistor are connected in series to a battery of e.m.f. 3.0 V and negligible internal resistance. The graph shows the variation of potential difference of both thermistor and resistor to the current flowing in them.





If the resistance of the resistor is doubled, what is the e.m.f. of the new battery, assuming negligible internal resistance, required for current flowing in the circuit to remain the same?

A 4.0 V **B** 4.5 V **C** 5.0 V **D** 6.0 V